RESEARCH PROPOSAL

Carbon-Price Uncertainty and the Timing of Abatement Investment: Measuring Transition Risk in Hard-to-Abate Industry

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1. Motivation

Carbon pricing is the central instrument of industrial decarbonisation policy. The logic is straightforward: make emissions costly enough and abatement becomes the cheaper option, so firms invest. In the energy-intensive sectors where emissions are most concentrated, cement, steel, chemicals, refining, that investment has been persistently slower than the prevailing allowance price would appear to justify. The European Union Emissions Trading System (ETS) has operated for two decades, allowance prices have at times exceeded 80 euros (EUR) per tonne, and capture and fuel-switching technologies exist at commercial or near-commercial scale. Yet the observed rate of deep abatement investment in these sectors remains low.

The conventional explanation is that abatement remains too expensive: the marginal abatement cost curve sits above the carbon price for the relevant technologies. I think that explanation is incomplete, and that its incompleteness matters both for policy design and for how transition risk is measured. Abatement investment in these sectors is large, sunk and effectively irreversible. Capture equipment cannot be redeployed; a retrofitted kiln cannot be un-retrofitted. The payoff to that investment depends on a carbon price that is volatile, politically contingent, and subject to discrete regime change through allocation reform, market stability mechanisms, border adjustment and outright policy reversal.

Under irreversibility and uncertainty, the net present value rule is the wrong decision rule. A firm holding an irreversible investment opportunity with a stochastic payoff holds a real option, and exercising it destroys the option value of waiting. The standard result from the real options literature is that the investment trigger price exceeds the Marshallian threshold, sometimes by a wide margin, and that the wedge widens with volatility. Applied here, this yields a proposition with direct policy content: a carbon price high enough to make abatement profitable in expectation may still be too uncertain to trigger investment. Raising the mean price is not equivalent to reducing its variance, and a policy that raises both may achieve less than one that stabilises expectations at a lower level.

The same mechanism has a risk-measurement corollary. Delay is precisely the process by which a productive asset becomes a stranded one. If the option value of waiting keeps firms out of abatement through the period in which retrofitting remains economic, the eventual outcome is not orderly transition but asset write-down under a tightening cap. Transition risk assessments, including those now required of financial institutions under supervisory climate stress testing, typically model the carbon price path and its effect on cash flows. They rarely model the firm’s option to wait, which means they may misestimate both the timing and the distribution of losses.

2. Where this comes from

My MSc thesis addressed one instance of this problem. It examined the Peak Cluster carbon capture and storage project in the UK cement sector, modelling abatement costs by Monte Carlo simulation under uncertainty in capital expenditure, operating cost, capture rate and carbon price, and benchmarking the resulting cost distributions against UK Emissions Trading Scheme allowance prices. The framing was path dependency: cement combines chemically unavoidable process emissions, long-lived and geographically fixed capital, and a cost structure exposed to international competition, which together produce technological and institutional lock-in.

I expected to find that the technology cost curve was the binding constraint. It was not. Within plausible parameter ranges, whether capture investment cleared depended far more on the assumed credibility and trajectory of the carbon price than on the central cost estimate, and on whether the firms sharing transport and storage infrastructure could be assumed to commit together. The thesis could observe this; it could not establish it. A simulation study of one cluster cannot distinguish a genuine mechanism from a feature of the parameterisation, cannot test whether the pattern holds across firms and sectors, and cannot identify the effect of uncertainty separately from the effect of the price level. Those three limitations define this proposal.

3. Research questions

- How large is the option value of waiting created by carbon-price uncertainty, and how does the abatement investment trigger vary with volatility, mean reversion and the probability of policy regime change?
- Is there evidence, in installation-level emissions trading data, that carbon-price uncertainty delays abatement investment independently of the price level?
- Where abatement depends on shared infrastructure, how does strategic interaction between firms holding parallel options affect the timing of investment, and which instruments shift the equilibrium earlier?

4. Paper 1 — Optimal timing of abatement investment under a stochastic carbon price

The first paper develops a real options model of the abatement investment decision. A firm operates an emitting installation and holds the option to invest an irreversible fixed sum in abatement capital that reduces its emissions intensity. Its payoff is the avoided cost of allowances, which depends on a stochastic allowance price. The problem is one of optimal stopping: at what allowance price should the firm invest?

The modelling choices that matter are in the price process and the policy risk. Carbon allowance prices are not well described by geometric Brownian motion. Supply is administratively fixed and periodically revised, demand is driven by industrial output and fuel switching, and the market stability reserve introduces an explicit quantity feedback. A two-factor specification separating a short-run mean-reverting component from a stochastic long-run equilibrium level, in the tradition of Schwartz and Smith, is a more plausible starting point, and it has the useful property that the long-run factor can be interpreted as the market's belief about policy stringency. Policy regime change can then be layered on as a jump process governing discontinuous revisions to the cap or to free allocation.

The objects of interest are the investment trigger price, the wedge between that trigger and the static net present value threshold, and the comparative statics of that wedge with respect to volatility, speed of mean reversion, and jump intensity. I would also examine the effect of free allocation, which is not neutral: allocation based on production benchmarks alters the marginal payoff to abatement and can, under some rules, reduce the incentive to invest even as it protects competitiveness. Where closed-form solutions are unavailable, the model can be solved numerically, and this Department has particular depth in approximation methods for American-style options.

The intended contribution is to characterise the option value of waiting specifically attributable to carbon-price uncertainty, in a setting calibrated to real allowance market dynamics rather than a generic stochastic payoff, and to generate comparative statics that the second paper can test.

5. Paper 2 — Empirical test on installation-level emissions trading data

The second paper asks whether the mechanism modelled in Paper 1 is visible in data. The European Union Transaction Log records verified emissions, free allocation and surrender activity at installation level for approximately eleven thousand installations from 2005 to the present. It is the most granular long-panel record of regulated emitters anywhere, and it is public. Matched to allowance price series and to measures of price uncertainty derived from allowance futures and options, it permits a direct test of whether abatement responds to the level of the carbon price, to its uncertainty, or to both.

The identification problem is the obvious one: allowance prices are common across all installations at a point in time, so a simple regression of emissions intensity on price recovers nothing more than a time trend. Three sources of variation offer traction. First, exposure differs across installations because free allocation rules, benchmark values and carbon leakage designations differ by sector and were revised at known dates between trading phases, generating variation in the effective marginal price faced by otherwise similar installations. Second, abatement opportunity sets differ by technology and capital vintage, so the same price shock implies different option values across plants. Third, the introduction of the market stability reserve and, more recently, the carbon border adjustment mechanism, changed the perceived variance of future prices for some sectors more than others.

The empirical strategy would combine a difference-in-differences design exploiting staggered changes in allocation rules with a within-installation panel specification relating investment-like discontinuities in emissions intensity to contemporaneous price volatility. Because treatment timing is staggered and effects are likely heterogeneous, conventional two-way fixed effects estimators are known to be biased in this setting, and I would use heterogeneity-robust estimators alongside them. Anticipation is a genuine threat to identification here, since allocation reforms are negotiated publicly and in advance; event-study specifications with pre-trend testing are essential rather than decorative.

A complementary approach would treat observed abatement investments as realised option exercises and estimate the implied trigger price distribution directly, comparing it with the theoretical trigger from Paper 1. This is a structural rather than reduced-form strategy and is more demanding, but it speaks to the magnitude question that policy actually needs answered.

The contribution would be the first systematic estimate of how much carbon-price uncertainty, as distinct from the price level, has slowed abatement in the EU ETS — a quantity that matters for the design of contracts for difference, price floors and corridors, all of which are being actively debated.

6. Paper 3 — Coordination under shared decarbonisation infrastructure

The third paper addresses the feature of my thesis case that neither of the first two papers captures. Capture and storage economics depend on shared transport and storage networks whose unit costs fall steeply with throughput. An individual firm's payoff therefore depends on whether other firms in the cluster also invest. Each firm holds an option to wait, and each has an incentive to let others commit first — both to avoid bearing the fixed infrastructure cost alone and to learn from their experience. The result is a timing game in which the socially optimal outcome is simultaneous early investment and the equilibrium may be mutual delay.

The paper would model this as a game of optimal stopping with payoff externalities, characterising the conditions under which delay is an equilibrium and how the outcome varies with cluster size, cost asymmetry and the observability of rivals' actions. It would then assess candidate instruments: contracts for difference on the carbon price, capacity reservation agreements, public underwriting of transport and storage, and anchor-tenant arrangements. The question is not whether these instruments help but which margin each operates on — whether it reduces the payoff variance, the fixed cost, or the strategic incentive to wait.

Where the game becomes analytically intractable for realistic cluster structures, the structure of interdependence among decision criteria can be characterised empirically from expert judgement. I have applied DEMATEL for exactly this purpose in earlier work, constructing and normalising direct-relation matrices, computing total-relation effects and separating causal drivers from dependent outcomes. That is a complement to formal modelling rather than a substitute, but it is a defensible way to structure a problem whose relevant interdependencies are not directly observable in market data.

7. Data

- European Union Transaction Log: installation-level verified emissions, free allocation and surrender records, 2005 to present, publicly available.
- Allowance price and uncertainty measures: EUA spot and futures series, and implied volatility from allowance options where liquidity permits. UK ETS series for post-2021 comparison and as a natural experiment in market separation.
- Installation and firm characteristics: Orbis or comparable firm-level data matched to installations, recovering ownership, capital vintage, financial position and leverage.
- Policy parameters: benchmark values, carbon leakage lists, market stability reserve parameters and CBAM sectoral coverage, for constructing exposure measures and for calibration.
- Technology cost data: capture and fuel-switching cost estimates from engineering literature and project disclosures, for calibrating the investment cost distribution.

8. Methods, and an honest account of my preparation

The agenda requires continuous-time stochastic modelling and optimal stopping for Paper 1, panel econometrics with credible identification for Paper 2, and game-theoretic modelling for Paper 3. I want to be exact about where I stand relative to that, because a proposal that overstates its author's readiness is not useful to a supervisor.

What I have done: Monte Carlo simulation of cost distributions under multi-parameter uncertainty, benchmarked against a traded carbon price; DEMATEL multi-criteria decision analysis implemented in MATLAB, including matrix normalisation and total-relation computation; management of multi-round panel survey data and analysis in Stata through a randomised controlled trial; construction and validation of large gridded datasets for published work. My coursework includes econometrics, mathematics for economists, applied statistics, experimental design and data analysis, and modelling and simulation.

What I have not done: continuous-time stochastic processes, optimal stopping, or graduate-level econometric identification. My econometrics training is at bachelor level. This is the principal gap between my preparation and this agenda, and it is why the programme's core courses in probability and statistical inference, optimisation and microeconomics are the reason I am applying to this department rather than to a policy school. I am undertaking preparatory work in mathematics for economists ahead of any start date and expect the first year to be methodologically demanding.

9. Fit with the Department and indicative sequencing

The proposal sits within the Department's stated interests on both tracks: climate risk and predictive modelling in management science, risk analysis and the economics of natural resources in economics. Methodologically it belongs to the tradition of stochastic modelling of energy commodity prices and their derivative contracts, and of electricity spot and forward price dynamics in relation to market risk premia, which is where this Department's doctoral work in energy has been concentrated. Carbon allowances are a traded commodity with an established derivatives market and a price process shaped by regulatory design; the apparatus developed for power and commodity prices transfers to them, and comparatively little of it has been directed at the abatement investment decision itself.

Indicatively, the first year would be given to coursework and to developing Paper 1 alongside it, with the EUTL panel assembled in parallel. Paper 2 would occupy the second and third years, since the identification work is the most demanding component. Paper 3 would follow, drawing on whatever the empirical work reveals about which margins actually bind. I would expect the agenda to be reshaped in discussion with a supervisor, and I would be glad for it to be — the questions matter more to me than this particular route through them.

What I would add that is less common in this field is direct institutional knowledge of how carbon prices are produced. Through work on integrity criteria for carbon projects at SEforALL's Africa Carbon Markets Initiative, and on inventory and reporting review at the UNFCCC, I have seen how allowance supply, verification and allocation rules are negotiated. Those rules are not exogenous parameters; they are the object of sustained political contest, and they are where the policy component of transition risk originates. Building an exposure measure that reflects the mechanism rather than a stylised version of it is, I think, where the empirical contribution of this work will stand or fall.